

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GPFFfitDte200e400 = ld.fit_src_struct;
end

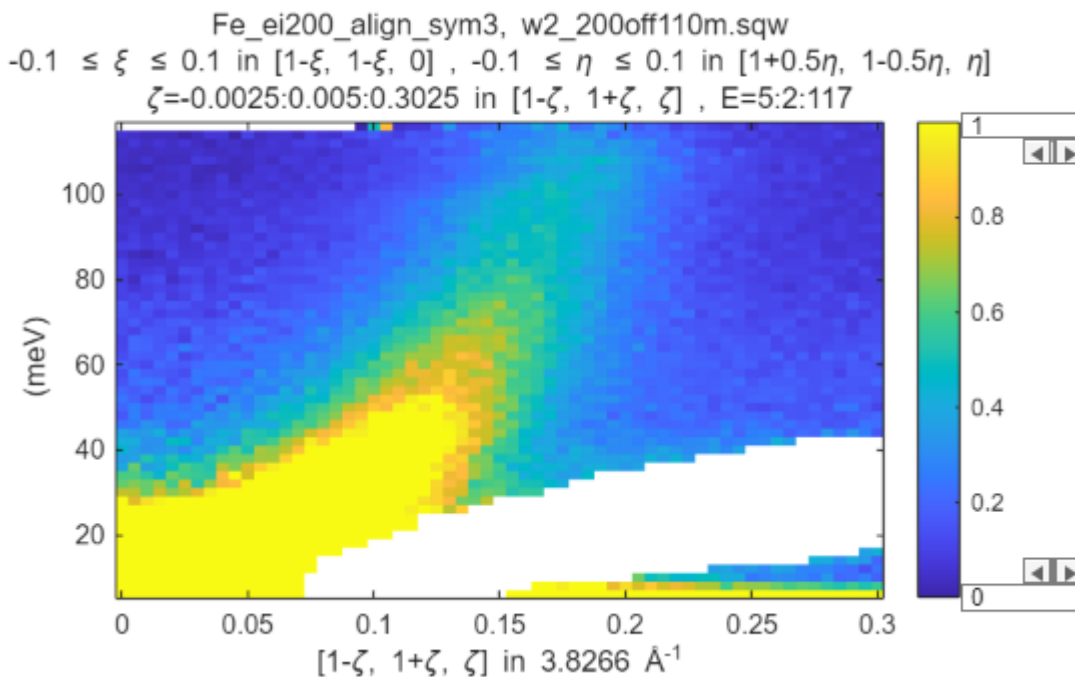
```

```

selectCuts= 'Ei200_off110GPmiff';
cuts_list = GP_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end

mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid', [0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



```

dir_name = "GP";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

```

```

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 110;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

```

```

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);

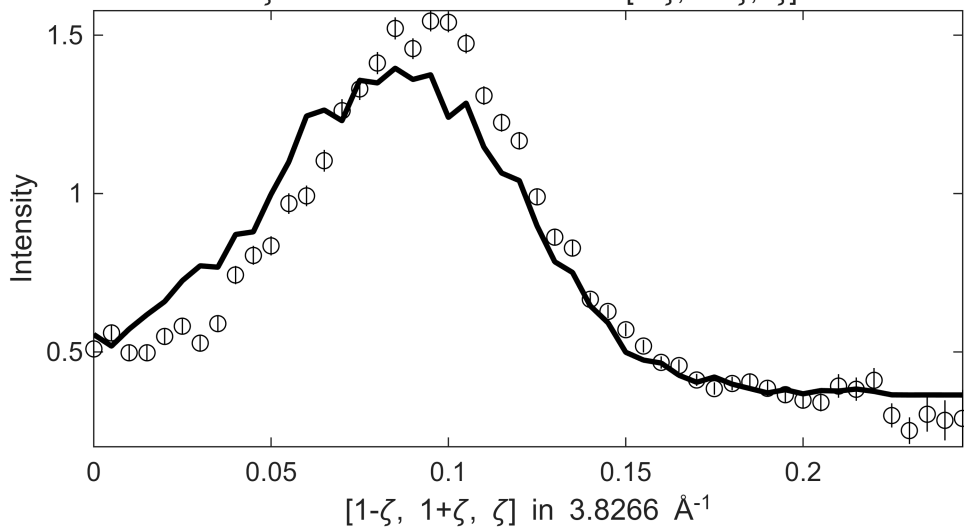
```

```

*****
**** En = 35±5 going to 105
*****

```

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S = 0.2 \pm 0.0095$; $J_0 = 7.8 \pm 0.01$; $\gamma = 0.0594991 \pm 0.00239924$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $30 \leq E$
 $\zeta = -0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$



```

*****
**** En = 45±5 going to 105
*****

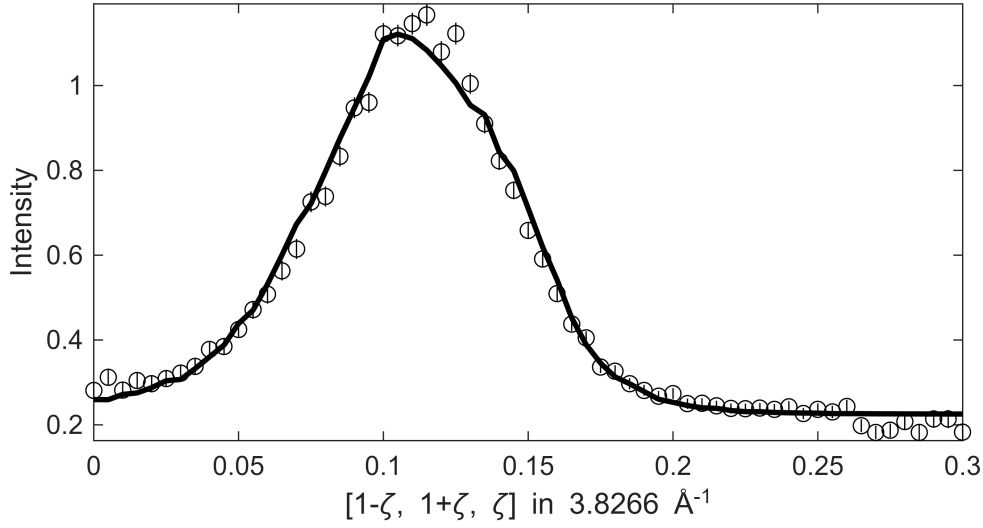
```

Fe_ei200_align_sym3, w2_200off110m.sqw

S=0.18±0.0027; J0= 6.6±0.02; gamma=0.556742±0.00916691

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 40 ≤ E

ζ=-0.0025:0.005:0.3025 in [1-ζ, 1+ζ, ζ]



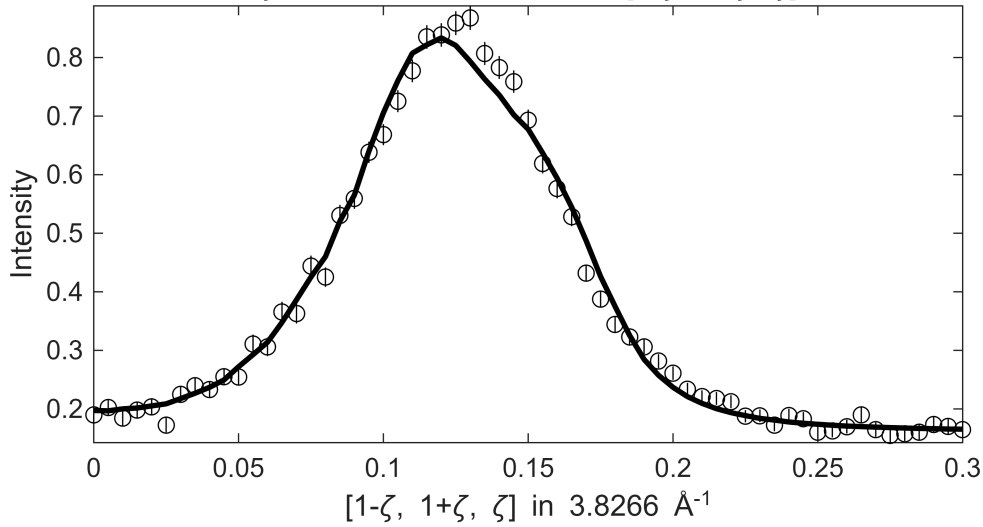
**** En = 55±5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw

S=0.18±0.0027; J0= 5.9±0.018; gamma=2.93555±0.0524335

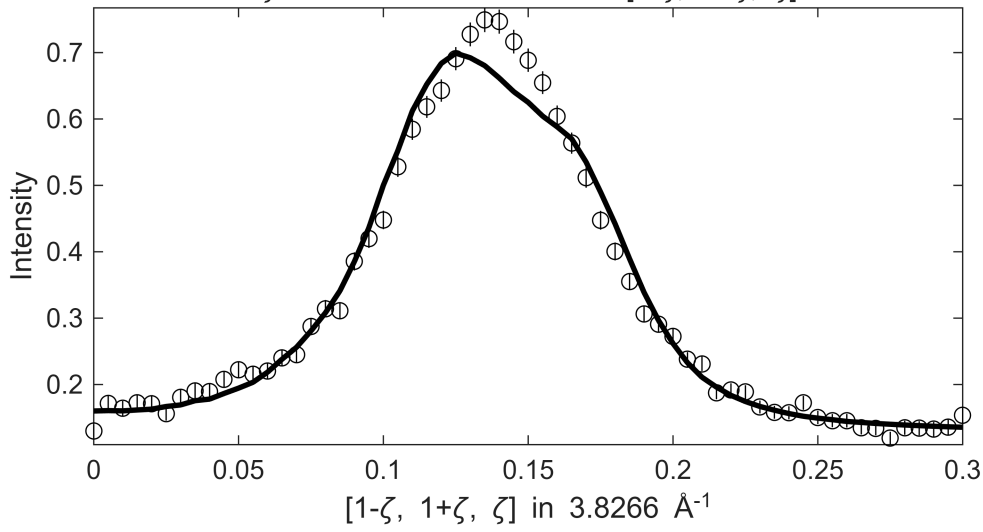
.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 50 ≤ E

ζ=-0.0025:0.005:0.3025 in [1-ζ, 1+ζ, ζ]



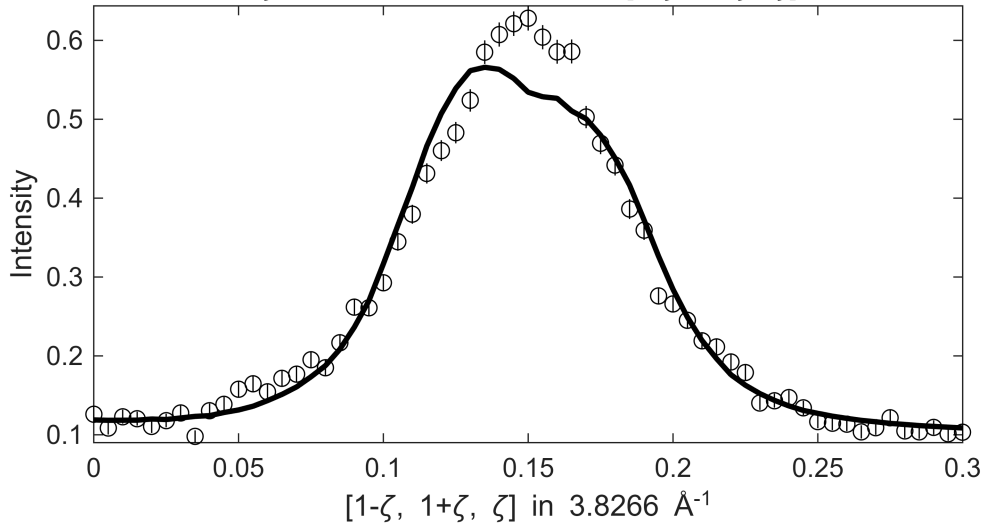
**** En = 65±5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.19\pm0.0036$; $J0= 5.6\pm0.019$; $\gamma=4.10467\pm0.100111$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $60 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$



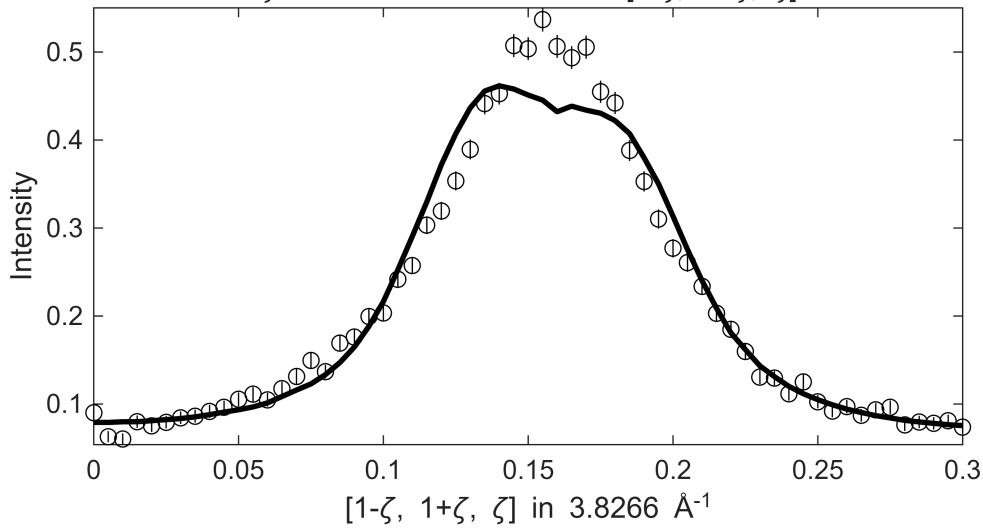
 **** En = 75±5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.18\pm0.0043$; $J0= 5.4\pm0.022$; $\gamma=5.3681\pm0.171564$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $70 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$



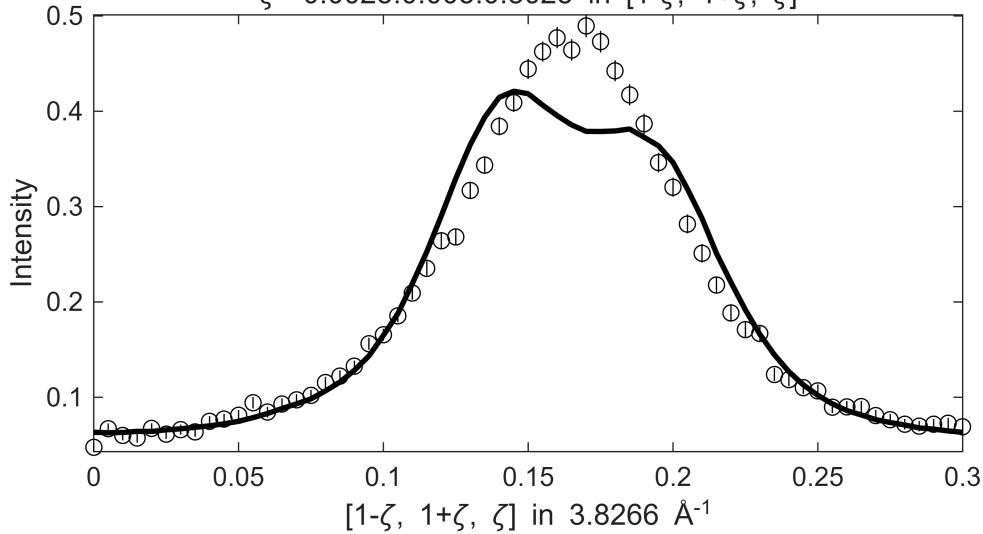
 **** En = 85±5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.21\pm0.0067$; $J0= 5.2\pm0.022$; $\gamma=7.03227\pm0.432047$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $80 \leq E$
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$



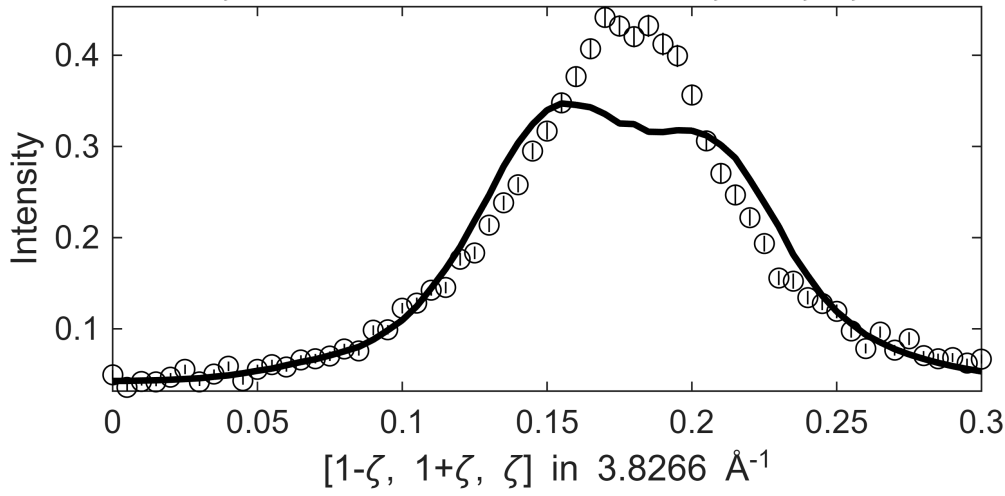
 **** En = 95 ± 5 going to 105

Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.23\pm0.011$; $J0= 4.9\pm0.025$; $\gamma=8.95638\pm0.939116$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $90 \leq E$:
 $\zeta=-0.0025:0.005:0.3025$ in $[1-\zeta, 1+\zeta, \zeta]$

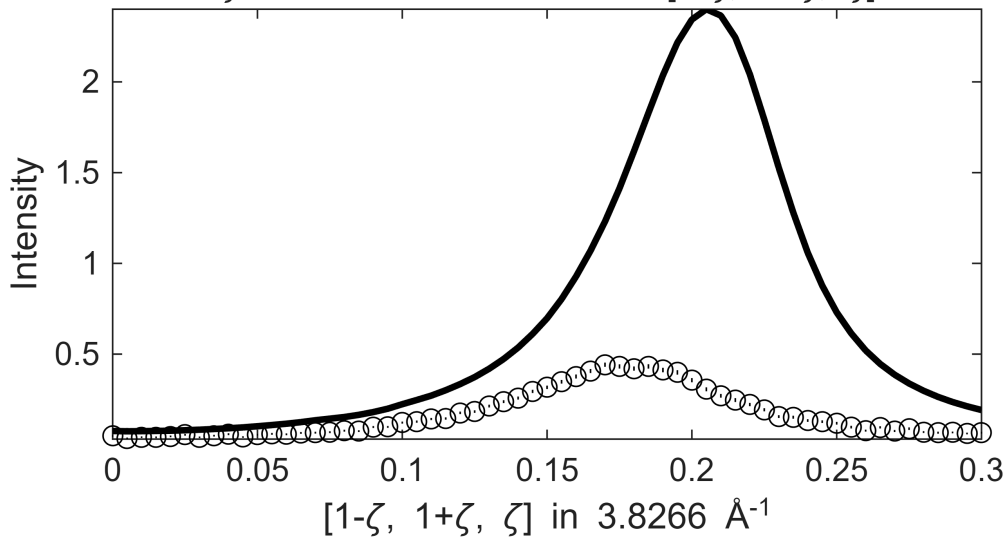


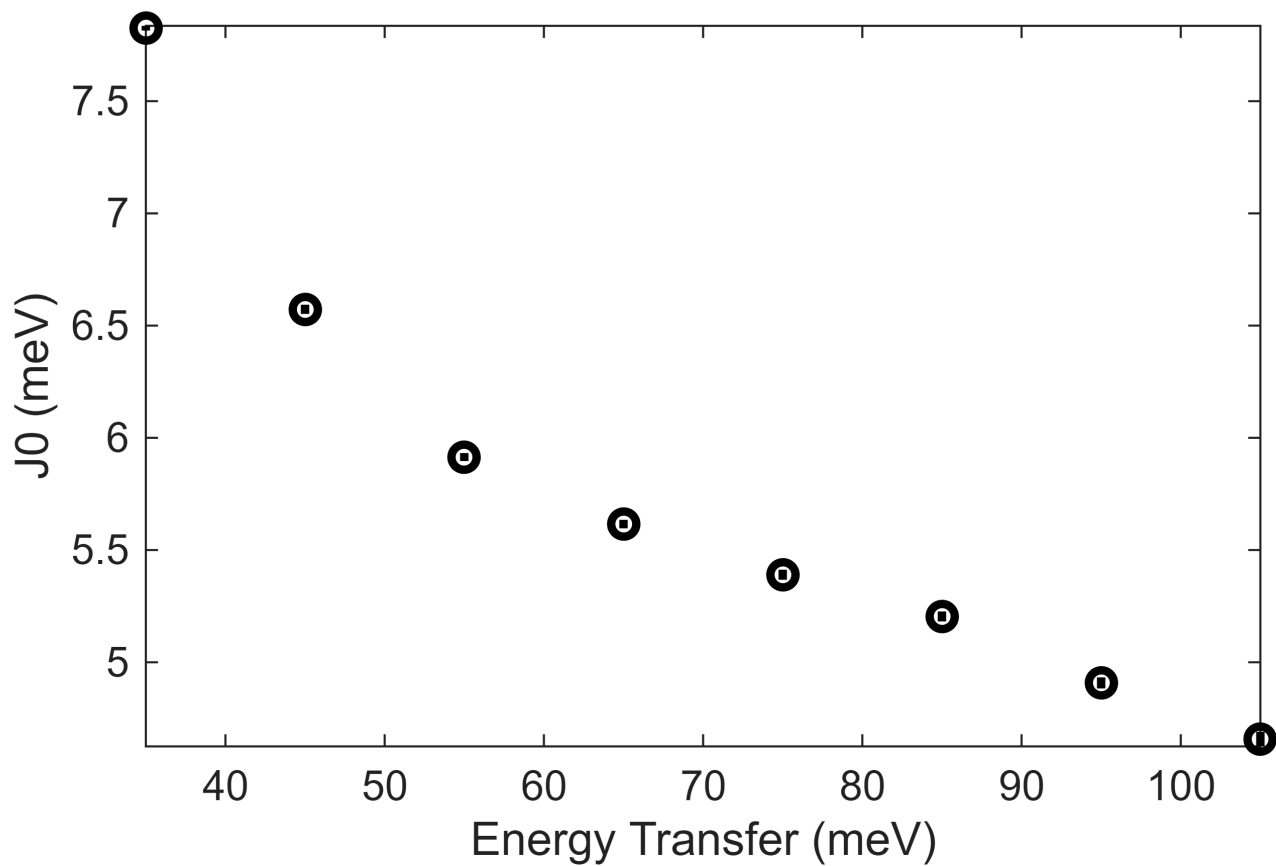
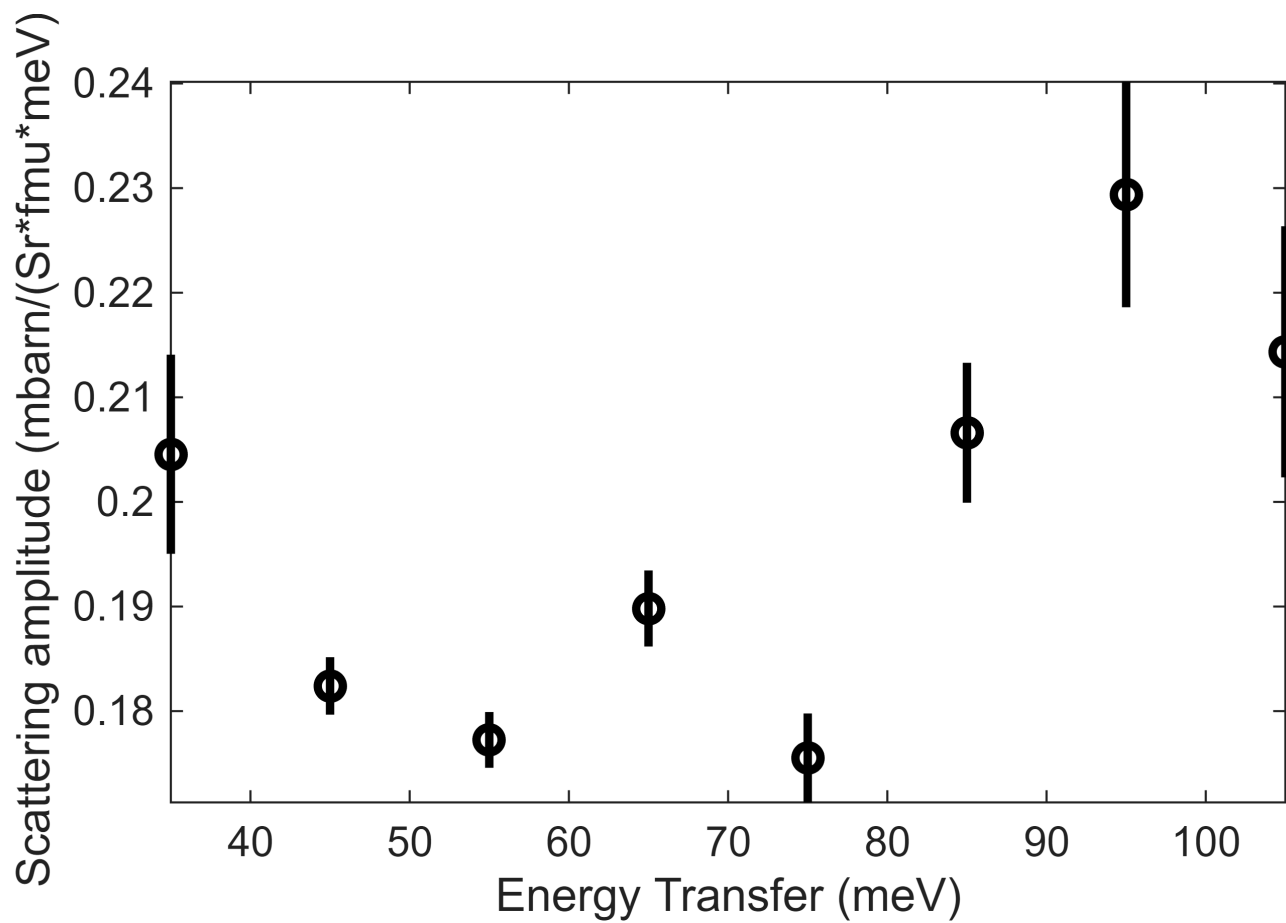
 **** En = 105 ± 5 going to 105

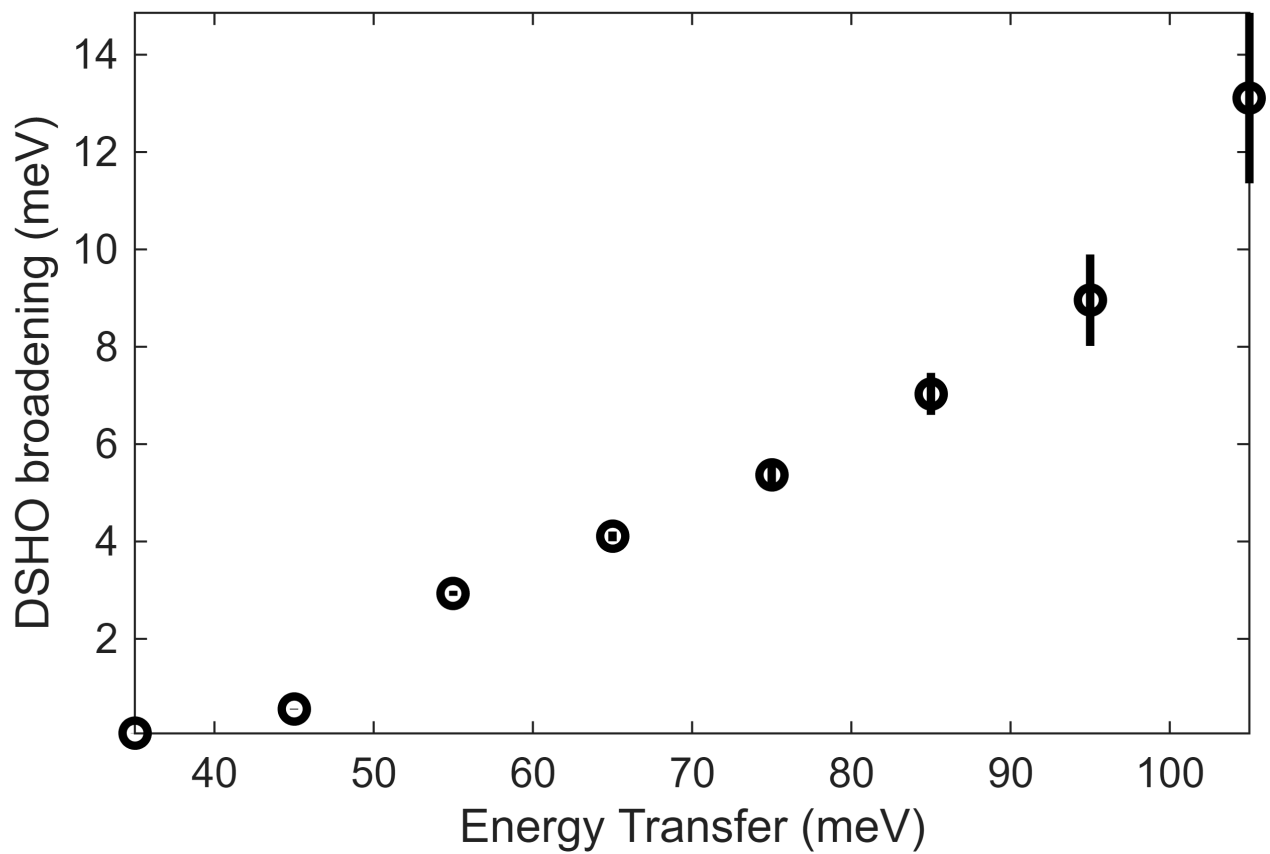
Fe_ei200_align_sym3, w2_200off110m.sqw
 $S=0.21\pm0.012$; $J_0=4.7\pm0.032$; $\gamma=13.1093\pm1.75068$
 $\xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $100 \leq \zeta \leq 1000$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off110m.sqw
 $\xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $100 \leq \zeta \leq 1000$ in $[1-\zeta, 1+\zeta, \zeta]$



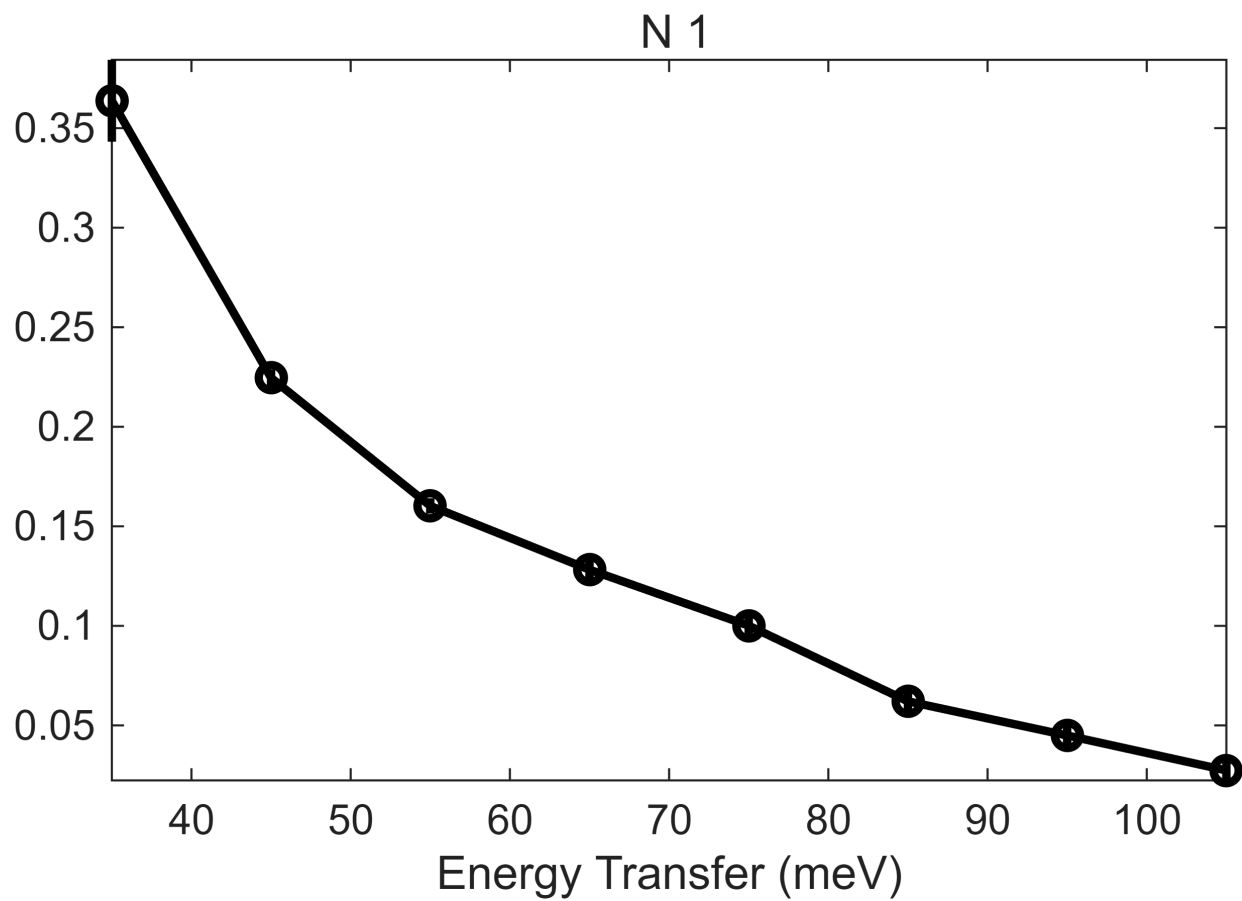




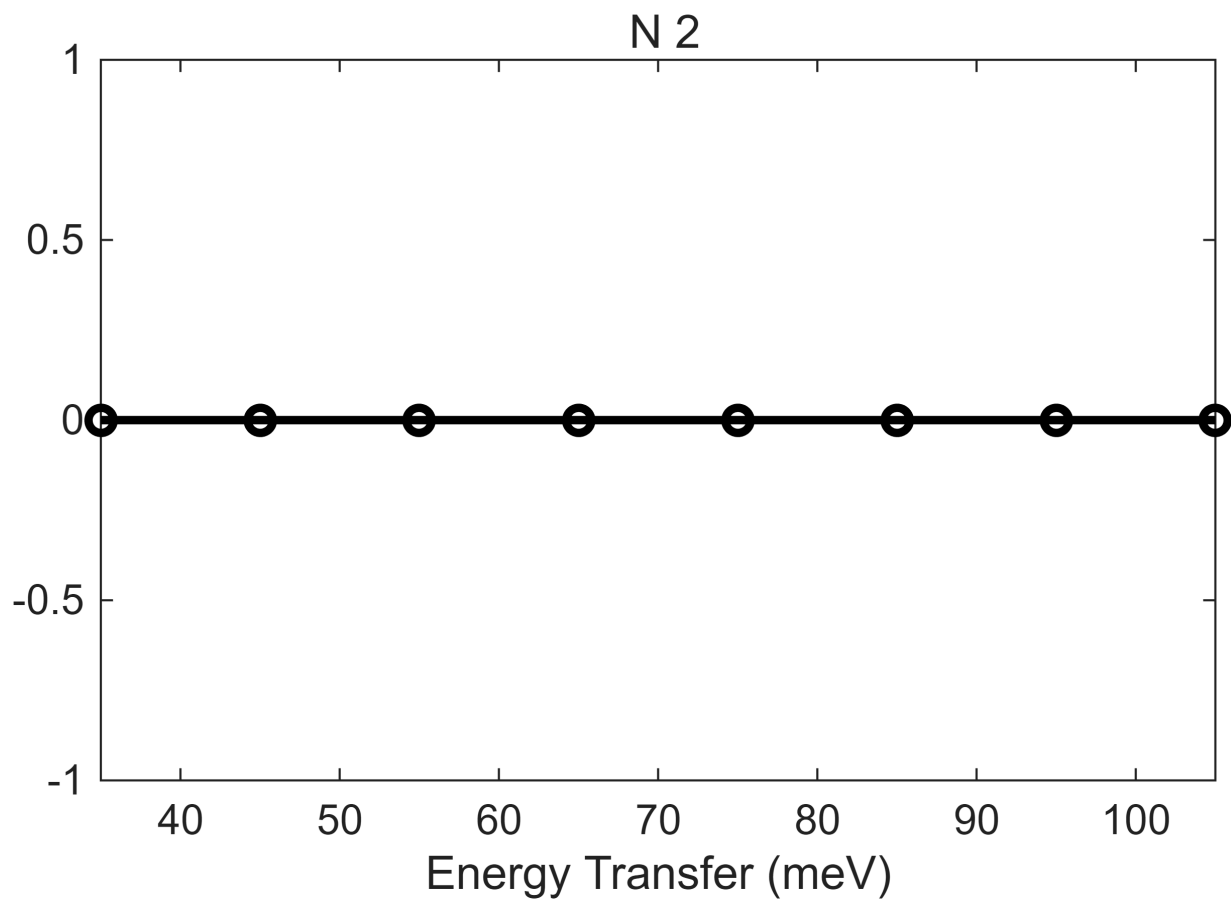
```
c1 = GP_FFCutsFits.(selectCuts);
GPFFfitDte200e400.(selectCuts) = [c1{:}];
```

```
beep
beep
beep
```

```
all_fit_par = GP_FFCutsFits.Ei200_off110GPmiff_all_fit_par;
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;
```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_  
figure;  
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_  
figure
```

```
fit_resEi200 = fit_res_rec;  
save(res_name,"fit_resEi200")
```